

Back to Entropy: Evaluating Base Language Models Beyond Perplexity and Benchmarks

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Abstract

Fixed task benchmarks expose important capabilities, but their model rankings can conflict and their accuracy scores discard most token-level evidence. Predictive cross-entropy uses every target token, yet ordinary perplexity cannot compare models with different tokenizers. We turn the training objective into a practical base-model selection protocol: normalize held-out loss by UTF-8 bytes, refresh the target-domain corpus, conservatively quarantine contamination-risk candidates, and measure every model before and after the same controlled adaptation procedure. We apply the protocol to 11 base models (0.5B–35B parameters) on news, Reddit, and Hacker News. On the marker-masked news evaluation, adapted BPB and HellaSwag have near-identical rankings (Spearman $\rho = 0.982$, seven of 11 exact ranks, maximum displacement one), as expected for two conditional-continuation measures. MMLU-Pro and GSM8K agree less ($\rho = 0.764$ and 0.727): Qwen models move upward, while Gemma models move downward, exposing family-specific knowledge, arithmetic, prompting, and answer-format effects that general text likelihood should not erase. Within each corpus, adaptation materially changes the zero-shot ordering ($\rho = 0.609$ – 0.682); after adaptation, rankings transfer strongly across corpora ($\rho = 0.991$, 0.955 , and 0.964). Diagnostic token- and context-level studies show that adaptation largely corrects surface-form calibration and retains its gain through 2,048-token inference. The release provides code, a versioned protocol, dataset fingerprints, auditable results, and a submission-ready leaderboard.

1 Introduction

For decades, language models were compared by predictive cross-entropy and its exponentiated form, perplexity. The metric was broad and statistically efficient—every next token contributed evidence—but token-level normalization made scores incomparable across vocabularies. Evaluation then shifted toward fixed task suites. MMLU-Pro, HellaSwag, and GSM8K are useful precisely because they isolate different capabilities [23, 24, 25]; the same specialization also makes their rankings disagree. Once benchmark questions become development targets or enter pre-training corpora, a high score may additionally reflect task preparation or overlap rather than transferable base-model quality [15, 19].

Recent evidence makes the return to loss timely. Cross-entropy can have higher signal and lower noise than accuracy [18], and train-before-test evaluations show that light, task-specific preparation substantially harmonizes benchmark rankings and restores their connection to perplexity [19]. The unresolved practical question is how to make that connection useful when choosing among off-the-shelf base models with different tokenizers for a new domain.

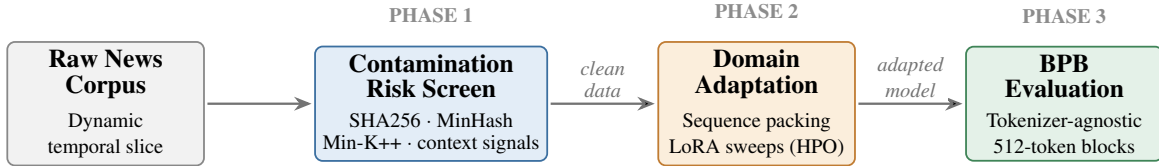


Figure 1: The standardized adapt-then-measure pipeline. Deduplication and contamination-risk screening (Phase 1) precede controlled adaptation (Phase 2) and marker-masked BPB evaluation (Phase 3).

We answer with a controlled *adapt-then-measure* protocol on evaluator-owned text. Bits per byte (BPB) places every model on one physical denominator. A recent target-domain snapshot limits staleness; internal deduplication and model-based risk signals trigger conservative quarantine before training or scoring; and a fixed LoRA configuration measures candidates under the same data and update budget. The endpoint, adapted BPB, answers “which base model best models this domain after equal preparation?” The reduction from zero-shot to adapted BPB answers a different question: “how much calibration did this corpus require?” Neither number is presented as a substitute for a task benchmark when the deployment task is already known.

Our contributions are: (1) a tokenizer-agnostic, fresh-data protocol for comparing adaptable base models; (2) a complete 11-model accuracy and rank audit on MMLU-Pro, HellaSwag, and GSM8K that explains where entropy rankings agree and diverge; (3) a 33-cell, three-domain experiment separating within-domain adaptation effects from cross-domain rank transfer; (4) token- and context-level diagnostics of what the adapters learn; and (5) a public implementation with versioned benchmark manifests, dataset fingerprints, machine-readable results, and a validated leaderboard. Code and artifacts are available at <https://github.com/DavidAfonsoValente/entropy-bench>.

2 Related Work

From perplexity to reliable loss evaluation. Language modeling was originally grounded in byte- and character-level entropy [1], while token-level perplexity depends on each model’s vocabulary and segmentation [2, 4]. Normalizing log-likelihood by UTF-8 bytes restores a common unit: The Pile used BPB for cross-tokenizer comparison [3], and Paloma standardized broad, domain-resolved measurement of language-model fit [5]. Heineman et al. evaluate 465 models over 30 benchmarks and show that loss-based metrics can offer better signal, noise, and scaling-law reliability than accuracy [18]. Our contribution is therefore not BPB itself; it is the operational protocol around the metric for comparing heterogeneous, off-the-shelf base models on new target data.

Why task rankings disagree. MMLU-Pro, HellaSwag, and GSM8K measure knowledge, conditional continuation, and mathematical problem solving under distinct elicitation and scoring rules [23, 24, 25]. Such scores can be indispensable for a known deployment task, but accuracy is discrete and benchmark-specific. Observational scaling work finds that a low-dimensional capability structure explains much public-model performance [20], while controlled model families exhibit a smooth relationship between validation loss and average downstream error [21]. We test that relationship across families and report both alignment and systematic residuals rather than collapsing the tasks into an uninterpreted macro-average.

Equal preparation. Dynamic evaluation updates a language model on recent test history to im-

prove predictive likelihood [26, 27]; domain-adaptive pre-training improves fit to a target distribution [28]. Closest to our model-selection motivation, Zhang et al.’s *train-before-test* gives 61 models identical supervised, benchmark-specific preparation across 24 tasks: average cross-benchmark Kendall agreement rises from 0.52 to 0.76, and pre-adaptation BPB predicts post-adaptation task potential for base models [19]. Their result motivates our design but the estimand differs. They rank post-fine-tuning task accuracy on public labeled benchmarks; we rank held-out language-model fit after unsupervised adaptation to a fresh target-domain corpus. We also standardize the byte denominator, screen contamination risk, and test rank transfer across raw-text domains.

Contamination controls. Models can memorize duplicated pre-training text [8, 9]; deduplication and contamination analysis are therefore necessary [10, 15]. Reference-free membership signals such as Min-K% and Min-K++ are useful screens [11, 12], but no black-box score proves that a sample occurred in a proprietary training set. We consequently use soft signals only for conservative quarantine and report them as risk indicators, not verified leaks.

3 The Evaluation Pipeline

The pipeline (Figure 1) is chronological: corpus construction and screening are completed before any sample can influence adaptation or scoring.

3.1 Phase 1: Deduplication and Contamination-Risk Screening

We first enforce exact split disjointness and remove exact and MinHash/LSH near-duplicates. For models whose pre-training corpora are unavailable, we then apply three reference-free screens:

- **Min-K++:** average the normalized likelihood of the lowest-probability $K = 20\%$ target tokens, using the model’s probability-weighted vocabulary mean and variance at each position; unusually high within-split outliers become candidates.
- **Context sensitivity:** compare target NLL with and without randomly sampled same-corpus prefix documents; unusually large confidence gains add a soft risk signal.
- **Perturbation preference:** for already flagged candidates, compare the original sequence with two word-swap/placeholder perturbations. Preference for the original triggers conservative quarantine, but is not treated as proof of memorization because fluency alone can produce the same outcome.

The screen runs per model and the union of candidate indices is quarantined globally. Thus every model receives the same retained corpus. This policy reduces overlap risk without making an unverifiable claim about membership in proprietary pre-training data.

3.2 Phase 2: Controlled Domain Adaptation

The retained corpus is packed into contiguous $L=512$ -token blocks along document boundaries. Every headline model–corpus cell uses LoRA rank 16, $\alpha = 32$, dropout 0.05, learning rate 10^{-4} , effective batch 32, all-linear targets, and 250 update steps. This fixes the data and update budget, although larger models still require more FLOPs and have more trainable LoRA parameters. Separately, the mechanistic diagnostics in Section 5.2 use three adapters selected by an exploratory multi-fidelity sweep; those runs explain the source of adaptation gains but do not determine the public ranking.

3.3 Phase 3: Tokenizer-Agnostic BPB Evaluation

Cross-entropy loss ($\mathcal{L}_{\text{nats}}$) aggregated over the held-out corpus is unified onto byte-space using the UTF-8 byte-length ($|B|$) of the source text:

$$\text{BPB} = \frac{\mathcal{L}_{\text{nats}}}{\ln(2)} \times \frac{|Tokens|}{|B|} \quad (1)$$

Zero-shot and adapted BPB are both measured on the held-out test split. Loss is accumulated over scored tokens and divided by the exact UTF-8 byte count represented by those targets. The packed test stream uses independent, non-overlapping 512-token blocks, so context resets at every cut. Injected document-start markers remain available as context but are always masked as prediction targets in the headline protocol. This point matters: an older exploratory run scored those markers; all core tables, correlations, the manifest, and the leaderboard use the corrected marker-masked rerun.

4 Experimental Setup

Models. We evaluate 11 base checkpoints spanning dense Transformers, one sparse MoE, and one Liquid architecture (0.5B–35B parameters); the cohort is listed in Tables 2 and 5.

Data. The primary snapshot is Google News dated 8 June 2026 (119,054 articles, ~361 MB of text). Reddit and Hacker News provide the transfer check in Table 1. Every corpus uses an 80/10/10 train/validation/test split (seed 42) with zero ID and normalized-text intersection across splits.

Static benchmarks. For Section 5.4, every base model is scored with *lm-evaluation-harness* 0.4.12 and vLLM 0.25.1 (bf16, greedy generation, tokenizer-native BOS). MMLU-Pro uses 5-shot chain-of-thought and exact match on one stratified 1,000-question subset (seed 42); HellaSwag uses 10-shot length-normalized multiple-choice accuracy on all 10,042 validation examples; GSM8K uses 5-shot chain-of-thought and flexible numeric extraction on all 1,319 test examples. These are directly observed scores under one harness, not vendor-reported model-card values.

Descriptive uncertainty and ranks. We orient every rank so 1 is best, compute Pearson correlation on score values, Spearman correlation on ranks, and Kendall agreement on pairwise order. Wilson 95% intervals show test-set sampling uncertainty for accuracy. They do not include run-to-run, prompt, subset-selection, or model-family uncertainty; the cohort is purposive, so all correlations are descriptive rather than population estimates.

Hardware. Adaptation ran under Slurm on the Leonardo (CINECA) GPU partition, 3 nodes $\times$ 4 NVIDIA A100 64 GB; the static-benchmark suite ran separately on A100 GPUs.

5 Results and Discussion

5.1 Core Result: Adapted Rankings Transfer Across Domains

We first test the claim needed for practical model selection: after the same light adaptation, does the ordering identify broadly strong base models rather than one news-specific winner? We prepare three

corpora with the same grouping and cleaning recipe—exact deduplication, Gopher quality heuristics, and MinHash/LSH near-duplicate filtering at Jaccard 0.60—and run all 11 models on every corpus.

Corpus	Raw records	Documents	Median words	Text
Google News, 8 Jun 2026	119,054	119,054	362	361 MB
Reddit, 6–10 Aug 2026 [†]	3,054,623	141,527	624	533 MB
Hacker News, 6–10 Aug 2026	60,814	6,086	690	46 MB

Table 1: Corpora in the 33-cell fixed-protocol experiment. Reddit and Hacker News comments are concatenated into document-length groups. [†]The Reddit source contains four available days in the stated range.

Model	News		Reddit		Hacker News	
	zero-shot	adapted	zero-shot	adapted	zero-shot	adapted
Gemma-4-31B	0.752	0.624	1.062	0.913	0.962	0.805
Gemma-4-12B	0.866	0.653	1.172	0.942	1.063	0.844
Qwen-3.5-35B-MoE	0.707	0.678	1.008	0.963	0.866	0.836
Ministral-3-14B	0.718	0.679	1.008	0.947	0.876	0.835
Qwen-3.5-9B	0.740	0.708	1.039	0.983	0.894	0.860
Qwen-2.5-7B	0.758	0.717	1.051	0.991	0.922	0.882
Qwen-3.5-4B	0.782	0.749	1.080	1.020	0.933	0.898
Llama-3.2-1B	0.811	0.771	1.109	1.055	1.011	0.976
Qwen-2.5-1.5B	0.833	0.797	1.128	1.072	1.002	0.967
LiquidAI-LFM2.5	1.340	0.852	1.779	1.149	1.687	1.051
Qwen-2.5-0.5B	0.935	0.901	1.222	1.161	1.099	1.064

Table 2: Marker-masked BPB under one fixed adaptation protocol. Lower is better. Rows are ordered by adapted news BPB; all eleven models occur in all three corpora.

Adaptation changes the choice. Zero-shot and adapted rankings are only moderately aligned: Spearman $\rho = 0.655$ on news, 0.609 on Reddit, and 0.682 on Hacker News, with maximum movements of seven, seven, and five places. Gemma-4-31B moves from fourth, fifth, and sixth zero-shot to first adapted; Gemma-4-12B moves from ninth in every zero-shot ranking to second, second, and fourth. Adapted BPB is therefore not a rescaling of ordinary held-out loss; it estimates fit after equal target-domain preparation.

The adapted order transfers. News and Reddit agree on nine of eleven exact ranks (Spearman $\rho = 0.991$; maximum displacement one). News versus Hacker News yields $\rho = 0.955$ and Reddit versus Hacker News $\rho = 0.964$, with maximum displacement two. Gemma-4-31B ranks first on all three corpora. Figure 2 shows that the result is not only ordinal: Pearson score correlations are 0.992, 0.977, and 0.990.

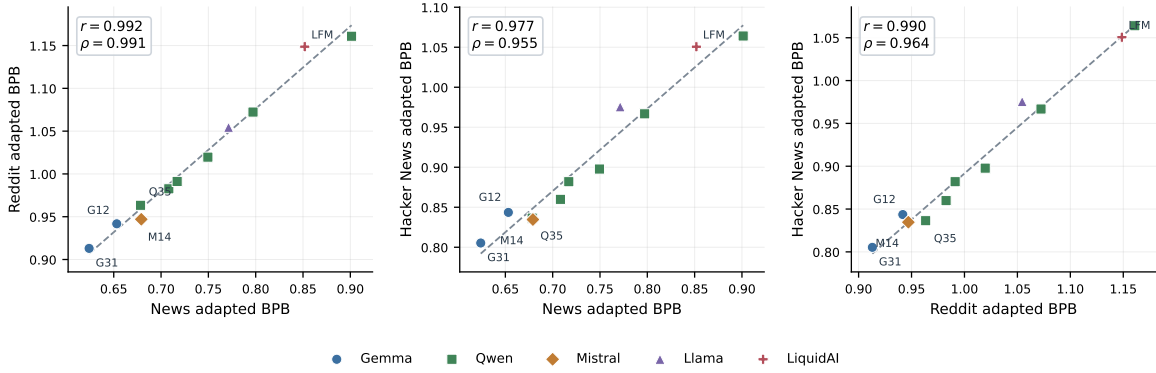


Figure 2: Pairwise adapted-BPB agreement across domains. Absolute difficulty changes, but points remain close to a monotone shared ordering. G31/G12 denote Gemma-4-31B/12B; Q35 is Qwen-3.5-35B-MoE; M14 is Ministral-3-14B.

Rank changes are near-tie changes. The three-model mid-size tier spans 0.0259 BPB on news but only 0.0089 on Hacker News; the two-model small tier spans 0.0258 and 0.0088. On Hacker News, Ministral-3-14B (0.835), Qwen-3.5-35B-MoE (0.836), and Gemma-4-12B (0.844) are separated by 0.009 BPB. A target-domain run should resolve choices inside such a tier; the cross-domain result supports the broader tier ordering, not an immutable total order.

Difficulty and order are distinct. Every model has its lowest BPB on news and highest on Reddit. Gemma-4-31B moves from 0.624 on news to 0.913 on Reddit, a 46% increase, yet remains first. Absolute BPB describes domain difficulty and attainable fit; relative rank describes which candidate is strongest under the shared adaptation protocol.

5.2 Mechanistic Diagnostics: What Adaptation Changes

To interpret the reduction, we score held-out blocks under each base model and an exploratory hyperparameter-selected LoRA adapter, then decompose the per-token log-probability gain. These three models span the observed range. This diagnostic run predates the marker-masking correction and is reported only to localize gain; it does not supply any headline BPB or leaderboard rank.

Model	Loss (nats/token)		Red.	Share of the gain		
	base	adapted		marker	next 16	remainder
Gemma-4-12B	2.838	2.015	29.0%	3.6%	5.0%	91.3%
LiquidAI-LFM2.5	4.298	2.728	36.5%	1.1%	1.9%	97.0%
Qwen-2.5-1.5B	2.737	2.647	3.3%	0.0%	0.0%	100%

Table 3: Exploratory token-level decomposition over 4,000 test blocks. *marker* is an injected document-start token, *next 16* is the post-boundary window, and *remainder* is every other position. Headline evaluations mask the marker target.

Surface-form calibration. Gemma-4-12B’s largest movers are <bos>, punctuation, paragraph breaks, and common function words; LFM-2.5 shows the same pattern. Across the top 200 gaining tokens, formatting, punctuation, and function words account for 83% of the gain for both models, while content words account for 5.2% for Gemma and 14.0% for LFM. Qwen-2.5-1.5B begins closer to the corpus distribution and allocates more of its smaller gain to dates and figures. For Gemma, the marker and its next 16 tokens contribute $3.6\% + 5.0\% = 8.6\%$ of this exploratory gain. These observations explain the three selected adapters; they are not assumed to identify a universal mechanism for all 11 models.

Context reveals steady-state quality. Figure 3 resolves loss and gain by position in the 512-token block and normalizes every bucket to UTF-8 bytes. Over the first eight positions, Gemma-4-12B averages 2.566 bits/byte and Qwen-2.5-1.5B averages 1.490; by positions 256–510, Gemma reaches 0.717 and Qwen 0.773. The crossing therefore remains after byte normalization and is not explained by vocabulary size alone. Once context is available, the 12B model shows its stronger steady-state language modelling. The opening 32 positions account for 23.0% of Gemma’s total gain and 12.9% of Qwen’s, making context recovery a clear component of the calibration signal.

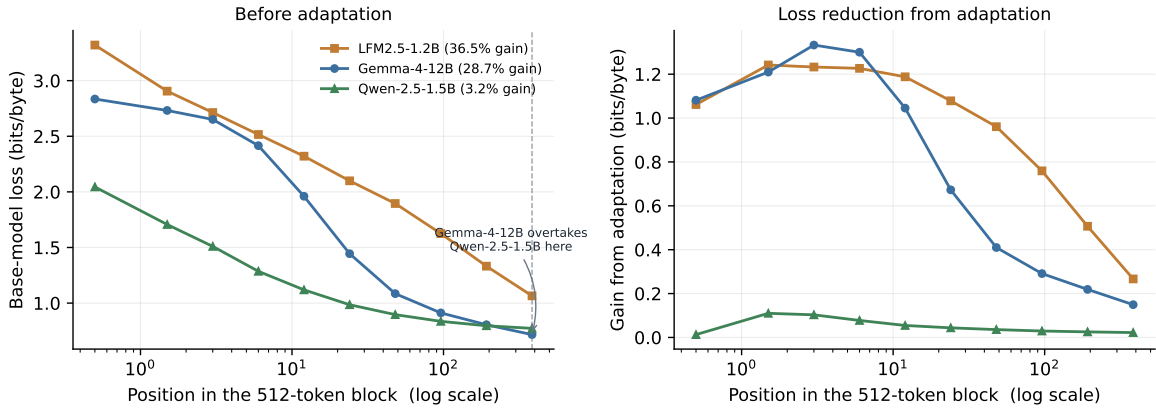


Figure 3: Exploratory base loss and adaptation gain by position in a packed 512-token block, in bits per UTF-8 byte. Positive gain is the loss reduction. The late-context crossing between Gemma-4-12B and Qwen-2.5-1.5B persists after byte normalization.

The 512-token adaptation transfers to longer inference contexts. We select 64 fixed held-out target windows and vary only the preceding context from 16 to 2,048 tokens, scoring the same target tokens at every length with exact byte normalization. Figure 4 shows that Gemma’s base BPB continues to recover beyond the training window, from 0.6465 at 512 tokens to 0.6054 at 2,048. Its adapted BPB is effectively unchanged ($0.5473 \rightarrow 0.5474$), so the adapter trained on 512-token blocks retains a 9.6% advantage at four times that context. Qwen’s smaller adaptation advantage also remains positive at 2,048 tokens ($0.7400 \rightarrow 0.7271$, a 1.7% reduction). Thus the base model continues to benefit from longer context, especially for Gemma, while the learned adaptation remains stable outside its training length.

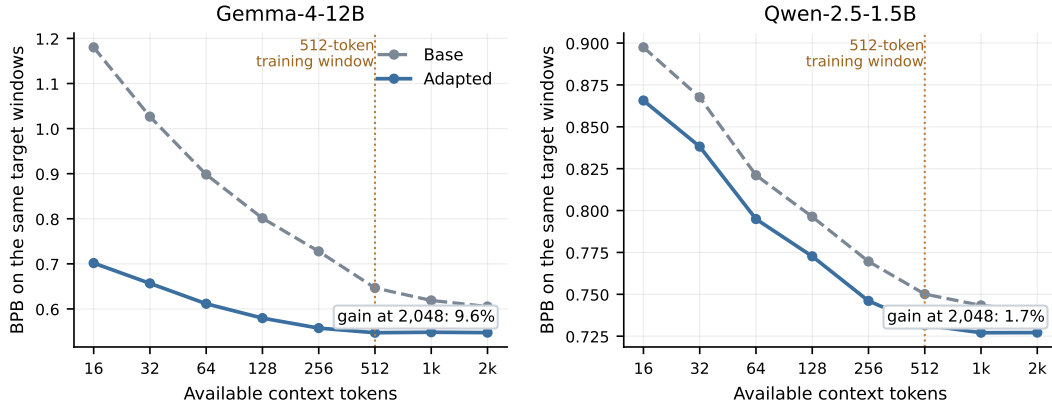


Figure 4: Fixed-target context-length diagnostic. Each point scores the same 64 held-out target windows; only preceding context changes. Both exploratory 512-trained adapters retain a positive gain through 2,048 inference tokens.

Interpretation. The adapted BPB level remains the selection metric. The zero-shot→adapted reduction is a calibration diagnostic; it must not be ranked as if a larger change implied a better endpoint.

5.3 Contamination-Risk Screen: What Was Removed

Table 4 reports the soft screening funnel on 1,000 sampled test sequences per model. It is a removal audit, not an estimate of each model’s memorization rate.

Model	Min-K++ flags	Context flags	Candidates	Original preferred	Quarantined
Gemma-4-31B	12	6	20	100%	20
Gemma-4-12B	11	6	11	100%	11
Ministral-3-14B	12	8	13	100%	13
Qwen-3.5-35B-MoE	6	8	11	100%	11
Qwen-3.5-9B	12	6	17	100%	17
Qwen-2.5-7B	10	6	14	100%	14
Qwen-3.5-4B	9	6	12	100%	12
Llama-3.2-1B	17	0	28	100%	28
Qwen-2.5-1.5B	14	4	17	100%	17
LiquidAI-LFM2.5	12	8	23	96%	22
Qwen-2.5-0.5B	8	2	13	100%	13

Table 4: Conservative screening funnel on the *test split* ($N = 1000$ sampled sequences per model; all splits are screened). “Original preferred” is preference over two synthetic perturbations, not task accuracy or proof of training membership. The global union removed 80 unique candidate sequences across splits and models.

The near-100% preference rates are expected because the distractors contain a word swap and a placeholder; a model can prefer the fluent original without having memorized it. We therefore make no scale–memorization claim from this table. The operational guarantee is narrower and auditable: all models use the same corpus after the union of flagged indices is removed.

5.4 Correlation with Static Benchmarks

We now compare the corrected adapted-news BPB with all three benchmark accuracies. Table 5 contains the complete 11-model intersection. Parentheses give within-column rank, always oriented so 1 is best.

Model	Adapted BPB ↓	MMLU-Pro ↑	HellaSwag ↑	GSM8K ↑
Gemma-4-31B	0.624 (1)	57.8 (2)	87.9 (1)	84.2 (3)
Gemma-4-12B	0.653 (2)	44.7 (7)	84.2 (3)	63.6 (7)
Qwen-3.5-35B-MoE	0.678 (3)	62.0 (1)	85.5 (2)	86.5 (1)
Ministral-3-14B	0.679 (4)	54.1 (4)	82.9 (4)	80.7 (5)
Qwen-3.5-9B	0.708 (5)	56.2 (3)	81.6 (5)	86.0 (2)
Qwen-2.5-7B	0.717 (6)	47.5 (6)	80.3 (6)	81.3 (4)
Qwen-3.5-4B	0.749 (7)	52.5 (5)	77.2 (7)	80.4 (6)
Llama-3.2-1B	0.771 (8)	11.7 (11)	65.8 (9)	6.5 (11)
Qwen-2.5-1.5B	0.797 (9)	25.0 (8)	67.8 (8)	61.1 (8)
LiquidAI-LFM2.5	0.852 (10)	15.5 (10)	60.9 (10)	55.0 (9)
Qwen-2.5-0.5B	0.901 (11)	16.7 (9)	51.8 (11)	34.7 (10)

Table 5: Directly observed benchmark accuracy in percent, with rank in parentheses. Bold marks the best point estimate in each column; it does not assert a statistically resolved lead. Adapted BPB comes from the marker-masked fixed-protocol news run.

	Pearson r	Rank ρ	Kendall τ	Exact ranks	Mean shift	Max shift
MMLU-Pro	−0.826	0.764	0.564	3/11	1.64	5
HellaSwag	−0.975	0.982	0.927	7/11	0.36	1
GSM8K	−0.614	0.727	0.564	0/11	2.00	5

Table 6: Alignment with adapted BPB. Pearson correlates BPB with accuracy, hence negative is agreement. Spearman and Kendall compare best-to-worst order and are oriented positive. “Shift” is absolute rank displacement.

HellaSwag and BPB measure the same local object. HellaSwag selects a continuation by length-normalized conditional log-likelihood, whereas BPB aggregates conditional log-likelihood over every target token. Their rank agreement is consequently 0.982: seven models keep their exact rank and none moves more than one place. Gemma-4-31B leads both (adapted BPB 0.624, HellaSwag 87.9%). This is convergent validity for conditional language modeling, not a claim that either metric measures every capability.

Knowledge and elicitation. MMLU-Pro agreement remains substantial but weaker ($\rho = 0.764$). The Qwen-3.5-35B-MoE leads at 62.0% and moves from BPB rank 3 to MMLU rank 1; Qwen-3.5-9B moves from 5 to 3. Gemma-4-12B moves from BPB rank 2 to MMLU rank 7. Unlike HellaSwag’s likelihood comparison, our MMLU-Pro protocol generates 5-shot chains of thought and requires an

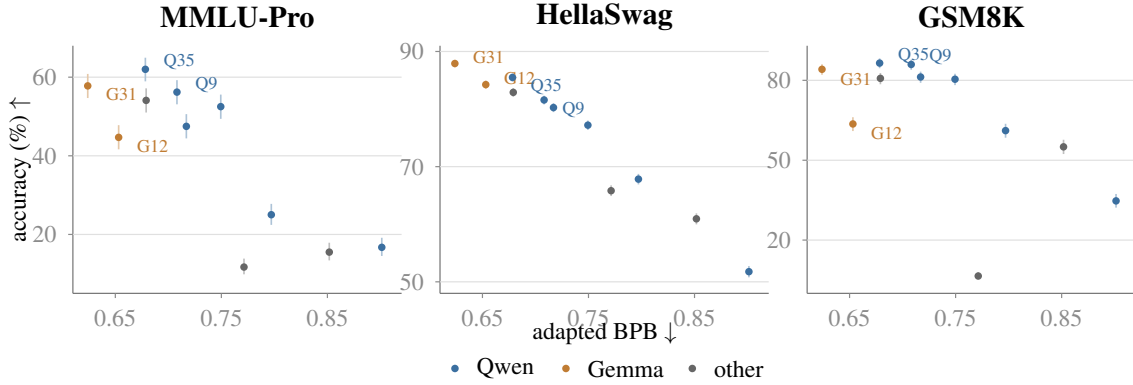


Figure 5: Benchmark accuracy versus corrected adapted-news BPB. Vertical bars are Wilson 95% intervals for finite test-set sampling only. Labels mark the four models central to the rank differences (G31/G12: Gemma; Q35/Q9: Qwen).

extractable exact answer. The residual therefore mixes factual coverage, reasoning, few-shot pattern continuation, and answer formatting.

GSM8K exposes a family-specific residual. No model keeps the same rank. Every Qwen model is unchanged or improves relative to BPB; both Gemma models decline, with Gemma-4-12B moving from rank 2 to 7. The contrast is large even among small models: Qwen-2.5-1.5B reaches 61.1% GSM8K at BPB rank 9, while Llama-3.2-1B reaches 6.5% at BPB rank 8. This supports a narrow interpretation: arithmetic data, chain-of-thought preparation, and generation-format compatibility vary systematically by model family. Because architecture, data mixture, token budget, and release generation are confounded, the experiment does not identify which cause dominates.

Point-estimate leaders are not equally resolved. HellaSwag’s leader exceeds the runner-up by 2.41 percentage points (independent-binomial $z = 5.03$), and their Wilson intervals do not overlap. MMLU-Pro’s 4.2-point top gap has $z = 1.92$ and overlapping intervals; GSM8K’s top two Qwen models differ by only 0.53 points ($z = 0.40$). Boldface therefore identifies the observed maximum, not a universal or necessarily significant winner. The repo publishes the numerators, denominators, intervals, ranks, and derivation script in `results/static_benchmark_analysis.json`.

6 Reporting Standard and Limitations

The evidence supports a practical reporting contract and also fixes the boundary of the claim.

Version the full estimand. A leaderboard row is comparable only when corpus revision, split, context length, marker masking, risk screen, adaptation configuration, update budget, and BPB aggregation match. Any change creates a new benchmark ID. The public leaderboard validates these fields rather than accepting an isolated number.

Separate level, change, and uncertainty. Adapted BPB ranks endpoint fit; zero-shot→adapted reduction describes calibration. Task accuracies retain their own denominators and intervals. A large reduction, bold point estimate, or tiny numerical BPB gap should not be promoted into a stronger claim than the underlying comparison supports.

Scope of model selection. Adapted BPB measures base-model potential after the specified unsupervised domain intervention. Zero-shot BPB better describes immediate language-model fit, and a task benchmark remains the relevant criterion when the deployment task and elicitation protocol are known. The framework complements rather than replaces capability and safety evaluation.

Limitations. The cohort contains only 11 English-centric base models and confounds architecture, data mixture, token budget, release generation, and parameter count. Fixed update and data budgets are not equal-FLOP budgets. Accuracy intervals cover test-set sampling only; static scores can change with prompts, harness versions, and extraction. The three diagnostic adapters and 64-window context study are explanatory probes, not cohort-wide replications. Reference-free contamination signals justify conservative removal but cannot verify proprietary training membership. Finally, the news text is third-party material: the release publishes its exact record-level fingerprint and verification tooling, but not the raw text without redistribution permission. These constraints limit causal and universal claims while leaving every reported comparison auditable.

7 Conclusion

The old choice between incomparable perplexities and narrow task accuracies is unnecessary. A recent raw-text snapshot, a byte denominator, conservative risk screening, and a fixed adaptation protocol turn predictive loss into an auditable base-model selection experiment. In our 11-model study, adaptation changes which models look strongest, yet the adapted order transfers across three domains.

The benchmark residuals make the result more useful, not less. HellaSwag almost reproduces the BPB order because both measure conditional continuation. MMLU-Pro and GSM8K reorder families because they additionally require knowledge, arithmetic, few-shot elicitation, generation, and answer extraction. The resulting recommendation is two-layered: use adapted BPB on fresh target data to choose broad base-model potential and stable tiers; use deployment-matched task and safety tests to resolve capability requirements and close calls.

Relative to train-before-test, the distinction is labels and estimand: it harmonizes ranks after supervised task adaptation; we compare off-the-shelf bases after unlabeled target-text preparation, across tokenizers. Released protocol checks make each leaderboard value traceable.

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